

6.2 Solving Compound Inequalities

Lesson Introduction

 Benchmark	MA.912.AR.2.6 Given a mathematical or real-world context, write and solve one-variable linear inequalities, including compound inequalities. Represent solutions algebraically or graphically.		 Learning Targets	- I can solve compound one-variable inequalities. - I can graph the solution of an inequality on a number line. - I can interpret solutions to problems as viable or non-viable options.	
 Benchmark Coherence	Building On		Working Toward		
	MA.8.AR.2.2 Solve two-step, linear inequalities in one variable, and represent solutions algebraically and graphically.		MA.912.AR.3.3 Given a mathematical or real-world context, write and solve one-variable, quadratic inequalities over the real number system. Represent solutions algebraically or graphically. MA.912.AR.4.2 Given a mathematical or real-world context, write and solve one-variable, absolute value inequalities. Represent solutions algebraically or graphically.		
 Essential Questions	- How do you solve compound inequalities? - How does the graph of the solution to a compound inequality look on a number line?				
 Lesson Narrative	In this lesson, students are introduced to compound inequalities. Students begin by exploring solutions to look for values that satisfy two different inequalities. This exploration leads into the guided instruction introducing compound inequalities. Additionally, students will learn about special cases that exist with compound inequalities before practicing with compound inequalities representing real-world contexts.				
 Component Summary	6.2.1 Warm-Up (~5 min)	Bell work on solving and graphing one-variable inequalities (<i>SE p. 298</i>)	6.2.4 Collaboration (5-10 min)	Special cases in compound inequalities Collaboration activity (<i>SE p. 301</i>)	
	6.2.2 Exploration (5-10 min)	Explore solutions to inequalities (<i>SE p. 298</i>)	6.2.5 Guided Practice (10 min)	Practice solving compound inequalities in real-world contexts (<i>SE p. 302</i>)	
	6.2.3 Guided Instruction (10 min)	Introduction to compound inequalities (<i>SE p. 299</i>)	6.2.6 Your Turn! (5-10 min)	Independent practice with compound inequalities in the real world (<i>SE p. 303</i>)	
	6.2.7 Wrap-Up (~5 min)	Error analysis activity (<i>SE p. 304</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Compound Inequality - Disjunction - Conjunction <u>Additional Terms:</u> N/A		N/A		N/A

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may forget to reverse the inequality symbol when multiplying or dividing by a negative. - Students may reverse the open and closed circle representations on the number line. - Students may confuse an “and” inequality with an “or” inequality when graphing.	- Students have not worked with compound inequalities prior to this point. - Consider using a timer or other method to help maintain your instructional timing. - If more time is needed for a particular component based on the needs of your students, consider making one or both of the following adjustments: Use the 6.2.2 as the Warm-Up, saving the 6.2.1 Warm-Up to use at another time. Consider assigning 6.2.6 Your Turn! as homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 6.2.4 Collaboration provides students with an opportunity to explore special cases in compound inequalities with a partner. The Think-Pair-Share routine is built into this activity with individual think time for some parts of each question, and discussion with a partner on other parts of the same question.	MA.K12.MTR.4.1: Discussion and Reflection 6.2.7 Wrap-Up provides students with an opportunity to complete an Error Analysis. During this activity, students will analyze the mathematical thinking of others, recognize errors, and suggest how to correctly solve the task.
 Differentiate Instruction	Consider using colored markers or highlighters to help students visualize inequality solutions when working with compound inequalities. For example, in 6.2.2 Exploration, in question 1e, students may need additional assistance with visualizing the solution to each individual inequality and then for both inequalities. By using different colors, students can see where the graph is independent and where they have a shared solution set.	
Additional differentiation opportunities are denoted throughout the lesson guide.		
Lesson Notes:		

6.2.1 Warm-Up: Bell Work

Component Overview: Students will complete a Bell Work activity where they solve and graph an inequality.



6.2.1 Warm-Up: Bell Work

1. Solve and graph the inequality.

$$\frac{2}{3}(12m + 93) \leq 86 - (-m - 4)$$

$$8m + 62 \leq 86 + m + 4$$

$$8m + 62 \leq 90 + m$$

$$7m \leq 28$$

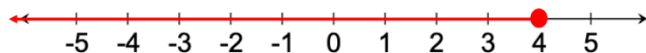
$$m \leq 4$$



This activity can serve as a formative assessment tool. Students' performance on this question can provide data for what students recall from the prior lesson about solving inequalities, and where in the process they may be getting stuck.



If timing is a concern for this lesson, consider reserving this Bell Work for another time and beginning the lesson with the next component.



6.2.2 Exploration: Is It a Solution?

Component Overview: Students will explore the concept of solutions to two independent inequalities, concluding with solutions that satisfies both inequalities. For this activity, students will need a partner.



6.2.2 Exploration: Is it a Solution?

1. Two inequalities are shown in the table.

- a. With your partner, discuss the similarities and differences of the two inequalities. Summarize your discussion.

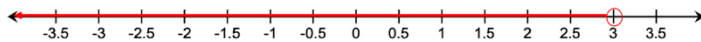
Answers will vary. Both inequalities have the variable f . Both inequalities require distribution. Inequality B uses the variable f on both sides. Inequality A uses the $<$ sign, while Inequality B uses the $\leq$ sign.

Inequality A	Inequality B
$-17.5 < -\frac{1}{2}(8f + 11)$ $35 > 8f + 11$ $24 > 8f$ $3 > f$	$-\frac{1}{2}(8f + 11) \leq 3f - 2$ $-4f - 5.5 \leq 3f - 2$ $-3.5 \leq 7f$ $-\frac{1}{2} \leq f$

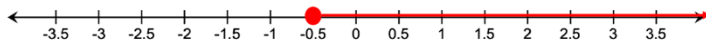
- b. Solve each of the inequalities in the space provided in the table.

Answers located in table above.

- c. Graph the solution set of Inequality A.



- d. Graph the solution set of Inequality B.



For question 1a, students should be able to identify some of the following:

- Inequality B has a variable on both sides, while Inequality A does not.
- The inequalities use the same variable.
- One side of each of the inequalities is the same expression.
- In one case, that same expression is greater than the other side of the inequality; in the other case, the expression is less than or equal to the other side of the inequality.

6.2.2 Exploration: Is It a Solution? (continued)

e. A list of values is shown.

-0.25	3	$\frac{3}{4}$	-84	1,000	$-\frac{4}{5}$
2.87	-1.4	0	18.6	$-\frac{1}{2}$	

With your partner, determine whether each value is in the solution set of Inequality A only, Inequality B only, or both. Record the values in the table below.

Solution to Inequality A Only	Satisfies Both Inequalities	Solution to Inequality B Only
-84	-0.25	3
$\frac{4}{5}$	$\frac{3}{4}$	1,000
$-\frac{4}{5}$	$\frac{3}{4}$	18.6
-1.4	2.87	
	0	
	$-\frac{1}{2}$	

f. Write two more numbers that are solutions to **both** Inequality A and Inequality B.

Answers will vary. 1 and 2. $1\frac{1}{2}$ and 2.7.

g. Describe the set of numbers that satisfies both Inequality A and Inequality B.

In order to satisfy both inequalities, the value must be between $-\frac{1}{2}$ and 3, including the $-\frac{1}{2}$.



Consider using colored markers or highlighters to help students visualize inequality solutions when working with compound inequalities. In this activity, in question 1e, students may need additional assistance with visualizing the solution to each individual inequality, and then for both inequalities. By using different colors, students can distinguish where the solution sets overlap (solutions that satisfy both inequalities) and where they only satisfy one inequality or the other.

6.2.3 Guided Instruction: Compound Inequalities

Component Overview: Students will be formally introduced to the concept of compound inequalities, including solving and graphing.



6.2.3 Guided Instruction: Compound Inequalities



A **compound inequality** is a conjunction of two or more inequalities.

1. The inequalities from the Exploration can be expressed as the compound inequality shown.

$$-17.5 < -\frac{1}{2}(8f + 11) \leq 3f - 2$$

The solution set of the compound inequality is the set of all values that satisfies both inequalities.

- a. Select the compound inequality that represents the solution set.

- ☐ $-\frac{1}{2} < f \leq 3$
- ☐ $-\frac{1}{2} \leq f < 3$
- ☐ $3 < f \leq -\frac{1}{2}$
- ☐ $3 \leq f < -\frac{1}{2}$

- b. Discuss with your partner what you think the graph of the solution set of this compound inequality would look like.

Answers will vary. I think this will look like two inequalities. I think this will look like one big inequality.

- c. The graph of the compound inequality is the **intersection** of the two graphs from the Exploration. Graph the solution set of the compound inequality.



Consider using a vocabulary strategy during this activity. This can include the use of an anchor chart, note-taking guide, or Frayer Model. This can help students conceptualize the terms to better identify them for problem solving in the future.



In question 1a, why did the signs, less than and less than or equal to, switch spots when compared to the original inequality?

6.2.3 Guided Instruction: Compound Inequalities (continued)

The inequality from the previous question is an example of a **conjunctive** compound inequality.



In a **conjunction**, the solution set satisfies all of the inequalities and is the **intersection** of the solution sets. A **conjunction** often uses the word “and.”

Compound inequalities can also be **disjunctive**.



In a **disjunction**, the solution set is any value that is in any of the individual solution sets. A **disjunction** often uses the word “or.”

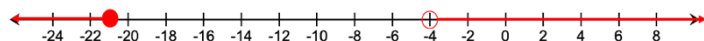
2. Consider the following example of a disjunctive compound inequality.

$$-3\left(1\frac{1}{2}m - 2\frac{3}{4}\right) < 22\frac{1}{4} - m \quad \text{or} \quad \frac{75m-9}{12} \leq -132$$

- a. Solve each inequality.

$-3\left(1\frac{1}{2}m - 2\frac{3}{4}\right) < 22\frac{1}{4} - m$ $-\frac{9}{2}m + \frac{33}{4} < 22\frac{1}{4} - m$ $-\frac{56}{4} < \frac{7}{2}m$ $-4 < m$	$\frac{75m-9}{12} \leq -132$ $75m - 9 \leq -1,584$ $75m \leq -1,575$ $m \leq -21$
--	---

- b. The solution set of the compound inequality is the set of all values that is a solution to either one of the inequalities. Graph the solution sets of **both** inequalities on the number line.



- c. Explain which of the following values is **not** a solution to the compound inequality.

-378 -5 19

-5 is not a solution because it does not satisfy either of the two inequalities.



A Frayer Model or a comparison chart for the terms “conjunction” and “disjunction” can be helpful for students. A common misconception is for them to confuse the two and apply what they’ve learned about each to the other.



- How can you tell, based only on the graph, which values are not part of the solution set?
- How do you know these values are not part of the solution set?

6.2.4 Collaboration: Special Cases in Compound Inequalities

Component Overview: Students will work with a partner or small group to explore special cases in compound inequalities.



6.2.4 Collaboration: Special Cases in Compound Inequalities

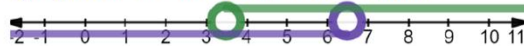
1. A disjunctive compound inequality is shown.

$$-22k + 5 < -14k - 23 \quad \text{or} \quad -4 < -3\left(k + \frac{1}{2}\right) + 17$$

- a. Solve each component of the compound inequality.

$-22k + 5 < -14k - 23$ $28 < 8k$ $3.5 < k$	$-4 < -3\left(k + \frac{1}{2}\right) + 17$ $-21 < -3\left(k + \frac{1}{2}\right)$ $7 > k + \frac{1}{2}$ $6.5 > k$
--	--

The graph of the solution set is shown.



- b. What do you notice about the graph of the two solution sets?

Answers will vary. I notice that the two graphs intersect. I notice that the entire number line is covered.

- c. Luz says that 3.5 is not a solution to the compound inequality because the first inequality is strictly “greater than.” Explain why Luz is incorrect.

Answers will vary. Luz is incorrect because 3.5 is a solution to the other inequality. Luz is incorrect because a solution only has to satisfy one of the inequalities, which it does.

- d. Discuss with your partner which of the following sets describes the solution set of the compound inequality. Explain your choice.

- $\{k: k \in \mathbb{N}\}$
- $\{k: k \in \mathbb{Q}\}$
- $\{k: k \in \mathbb{R}\}$
- $\{k: k \in \mathbb{Z}\}$

All real numbers satisfy at least one of the inequalities.



MA.K12.MTR.6.1: Reasonableness

This component provides an opportunity to incorporate MA.K12.MTR.6.1. For this activity, the goal is for students to understand situations where there could be no solution or all real numbers. This is a concept that a lot of students struggle with. Students will often get the correct answer mathematically but not go back and read their inequality statement to see if it makes sense. A strategy is to teach students to read their final answer out loud and look at their graphed solution to see if it is reasonable.



If students need help with the symbols used in set-builder notation, refer them to Unit 4 Lesson 1 where they were introduced to the symbols.

6.2.4 Collaboration: Special Cases in Compound Inequalities (continued)

2. A conjunctive compound inequality is shown.

$$-1 < -z + \frac{3}{4}(z - 16) < \frac{1}{2}z + 12$$

- a. Rewrite the compound inequality as two single inequalities and solve each one.

$-1 < -z + \frac{3}{4}(z - 16)$ $-1 < -z + \frac{3}{4}z - 12$ $11 < -\frac{1}{4}z$ $-44 > z$	$-z + \frac{3}{4}(z - 16) < \frac{1}{2}z + 12$ $-\frac{1}{4}z - 12 < \frac{1}{2}z + 12$ $-24 < \frac{3}{4}z$ $-32 < z$
--	--

- b. Explain whether any of the values shown satisfy **both** of the inequalities.

192.9	22	$-\frac{1}{3}$
-------	----	----------------

None of these values would satisfy both inequalities because they are not true when substituted.

- c. Describe the solution set of the compound inequality.

There are no solutions to this compound inequality.



A compound inequality could have no solutions or could have a solution of all real numbers.



- Why would none of these values satisfy both inequalities?
- Can you think of any number that would satisfy both of them?



For students who finish early, challenge them to come up with one mathematical and one real-world scenario for a compound inequality with no solution or a solution of all-real numbers.

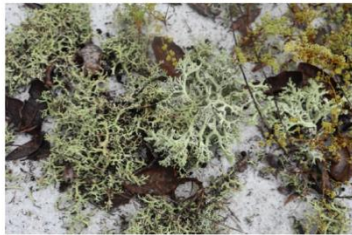
6.2.5 Guided Practice: Compound Inequalities in the Real World

Component Overview: Students will engage in a Guided Practice problem involving compound inequalities in the real world.



6.2.5 Guided Practice: Compound Inequalities in the Real World

1. *Cladonia perforata* is an endangered lichen found in the Atlantic Coastal Ridge, Lake Wales Ridge, Southwest Florida, and the North Gulf Coast of Florida. The habitat of this species is the white sand of the Florida scrub, already a rare and endangered type of ecosystem, and the lichen requires a specific spot within the habitat. To study the lichen after removing it from its ecosystem, botanists soak the lichen in water and then measure its surface area.



A botanist found that the lichen from Tiger Creek Preserve on the Lake Wales Ridge had surface areas that ranged between 431 square millimeters (mm^2) and 693 mm^2 . The botanist uses the inequality shown, where w is the weight in grams (g), to determine the weight range of the lichen.

$$431 \leq 524w + 85.16 \leq 693$$

- a. Solve the inequality.
 $345.84 \leq 524w \leq 607.84$
 $0.66 \leq w \leq 1.16$

- b. Check your solution with your partner.

- c. Complete the statement.

The lichens in Tiger Creek Preserve have a weight of between 0.66 and 1.16 grams.



At this point, students have not engaged with compound inequalities in real-world contexts. Consider leveraging a Three Reads routine to help students unpack the information from the problem, including important information and what it is asking.

6.2.5 Guided Practice: Compound Inequalities in the Real World (continued)

- d. Explain whether or not the *Cladonia perforata* in Tiger Creek Preserve could have a weight of 1 g.

Answers will vary. It could have a weight of 1g because it falls between the 0.66 and 1.16. It could have a weight of 1g because 1 is a solution to the inequality.



Have students share out their answers to questions 1d and 1e. This can help drive the conversation about solutions in context and what they represent.

- e. Explain whether or not the *Cladonia perforata* in Tiger Creek Preserve could have a weight of 0.6895 g.

Answers will vary. It could have a weight of 0.6895g because it falls between the 0.66 and 1.16. It could have a weight of 0.6895g because 0.6895 is a solution to the inequality.



For question 1e, students should walk away with an understanding that any mathematical solution is a viable solution because weight can have any number of decimal places. Since the mathematical solution does not include negative values, any solution is viable in this context.

- f. Explain why any solution to the mathematical inequality is a viable solution given the real-world context.

Answers will vary. Any mathematical solution is a viable solution because weight can have any number of decimal places, and since the inequality precludes negative numbers, any solution is viable.

6.2.6 Your Turn!

Component Overview: Students will complete a real-world, compound inequality problem independently.



6.2.6 Your Turn!

- The Homeland Security and Emergency Management Agency has emergency procedures in place for extreme temperatures in Washington, D.C.

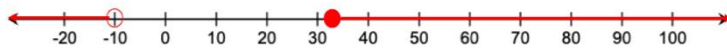
A temperature emergency is triggered if the temperature drops below 15° Fahrenheit ($^{\circ}\text{F}$) or reaches 92°F . The compound inequality shown, where c is the temperature in degrees Celsius, can be used to determine the temperature in Celsius that would trigger a temperature emergency in Washington, D.C.

$$\frac{9}{5}c + 32 < 15 \quad \text{or} \quad \frac{9}{5}c + 32 \geq 92$$

- Solve the compound inequality.

$$\begin{array}{ll} \frac{9}{5}c + 32 < 15 & \frac{9}{5}c + 32 \geq 92 \\ \frac{9}{5}c < -17 & \frac{9}{5}c \geq 60 \\ c < -\frac{85}{9} & c \geq \frac{100}{3} \end{array}$$

- Graph the solution set of the compound inequality. Approximate values as necessary.



- Complete the statement.

A temperature emergency is declared in Washington, D.C. if the temperature is less than $\underline{-9.4}$ $^{\circ}\text{C}$ or at least $\underline{33.3}$ $^{\circ}\text{C}$.



Consider using this time to pull a small group of students together who may have struggled with the previous activity. These questions are structured in such a way that it can be used as a miniature gradual release model with a small group.

Additionally, if time is a concern, consider assigning this component for homework.

6.2.7 Wrap-Up: Error Analysis

Component Overview: Students will complete an Error Analysis on solving a compound inequality.



6.2.7 Wrap-Up: Error Analysis

1. Simeon solved a compound inequality. His work is shown.

$$\begin{array}{lcl}
 18 < -3y + 72 < \frac{1}{3}(168y - 92) & & \\
 \swarrow & & \searrow \\
 18 < -3y + 72 & & 18 < \frac{1}{3}(168y - 92) \\
 -54 < -3y & & 18 < 56y - 46 \\
 18 > y & & 54 < 56y \\
 & & \frac{54}{56} < y \\
 & & \frac{27}{28} < y
 \end{array}$$

Simeon stated the solution set is $\frac{27}{28} < y < 18$.

Explain the mistake(s) Simeon made.

Answers will vary. When he rewrote the compound inequality as two separate inequalities, he used the wrong expression on the left-hand side of the second inequality. He should have used $-3y + 72$ instead of 18.



How students respond to this question can provide insight as to what students understand about compound inequalities.